

Yvann Le Fay, 2nd year PhD candidate in Statistics

Email yvann.lefay@ensae.fr
Github <http://github.com/ylefay>
Phone + 33 6 45 44 70 93
Date of Birth January 20, 2000
Citizenship Switzerland, France

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CREST - Office 3015
5 Av. Henry le Chatelier
Palaiseau, France
91120

EDUCATION

- 2024 - **Ph.D. in Statistics**, Institut Polytechnique de Paris France
Supervised by Pr Nicolas Chopin & Pr Matti Vihola (University of Jyväskylä).
Thesis: "Computational and Methodological aspects of Sequential Monte Carlo Algorithms."
- 2023 - 2024 **M.Sc. in Statistics, MVA**, École Normale Supérieure Paris-Saclay France
- 2020 - 2024 **M.Eng. in Statistics**, École N^{ale} de la Statistique et de l'Administration Economique (ENSAE) France

EXPERIENCES

- 2024 - Now **Teaching Assistant**, École N^{ale} de la Statistique et de l'Administration Economique (ENSAE) France
Courses: Stochastic processes, Monte Carlo methods, Statistical tests, Nonparametric estimation
Writing training course material, tutoring and grading students.
- 2023 **Research Assistant**, Aalto University Finland
5 month research internship, supervised by Dr. Adrien Corenflos and Pr Simo Särkkä.
Proposed the first probabilistic numerics solver for Stochastic Differential Equations (SDEs) in the additive-noise setting using Gaussian filtering methods.
- 2023 **Research Assistant**, National Institute for Research in Digital Science and Technology (INRIA) France
3 month research internship, supervised by Pr Demian Wassermann in the Parietal/MIND team.
Worked on hierarchical Bayesian models for medical imaging. Increased the test coverage of NeuroLang, a Python package for Probabilistic Logic Programming specialized to neuroimaging.
- 2022 **Quantitative Researcher**, Lombard Odier Investment Managers' (LOIM) - Multi-Asset Switzerland
6 month Off-cycle internship, supervised by Dr. Serge Tabachnik
Contributed to a proprietary scalable and highly modular framework for backtesting cross-asset trading strategies. Researched, implemented and backtested strategies (mainly momentum). Implemented data mining procedures and regression methods. Contributed to a statistics and reporting library. Received a renewal offer.
- 2021 **Data Scientist**, Société Générale (CDN, Retail Banking) France
3 month internship in data science and risk analysis, supervised by Pierre Bouché.
Backtested a credit decision tool, detected outliers in audit logs. Conducted a study on omnichannel customer behaviours as part of the fusion between Société Générale and Crédit du Nord.

PUBLISHED ARTICLES

- 1 **Y. Le Fay**, N. Chopin, and M. Vihola. On the complexity of standard and waste-free SMC samplers. *Preprint*.
- 2 **Y. Le Fay**, N. Chopin, and S. Barthelmé. Least squares variational inference. *Advances in Neural Information Processing Systems (NeurIPS, 2025)*
- 3 **Y. Le Fay**, A. Corenflos, and S. Särkkä. Modelling pathwise uncertainty of Stochastic Differential Equations samplers via Probabilistic Numerics. *Bayesian Anal. Advance Publication 1-24 (2025)*. DOI: 10.1214/25-BA1522

REFEERING FOR PEER-REVIEWED JOURNALS

Biometrika; Annals of Statistics

PRESENTATIONS IN SCIENTIFIC CONFERENCES AND SEMINARS

Scalable MCMC Sampling (only as an attendee, Zürich, Switzerland, 06/2026); Mostly Monte Carlo (Paris, 12/2025); NeurIPS (San Diego, USA, 12/2025); Recent advances in Machine Learning (Aussois, France, 11/2025); Sampling from the distribution (Télécom Paris, France, 11/2025); Algorithms & Computationally Intensive Inference seminars (University of Warwick, UK, 10/2025); BAYSM (University of Connecticut, USA, 11/2023, online).

PROGRAMMING SKILLS

Languages: Python, Ocaml, Wolfram Language

Technical stack: Linux, Git

Libraries and others: JAX

SOME REPOSITORIES

Least squares variational inference (in JAX, NeurIPS25)
<https://github.com/ylefay/LSVI>

Fast Bayesian SDE Solver (in JAX, Bayesian Analysis)
<https://github.com/ylefay/bayesianSDEsolver>